Electronic Supplementary Materials

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Title: Co-creating intergenerational knowledge about Hawaiian moi (*Polydactylus sexfilis*) reproductive cycles in a changing climate

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Supplementary 1: Hawaiian Language Glossary

Other than group names, definitions are drawn from Wehe²wiki² Hawaiian Language Dictionaries with slight variations. As Hawaiian words have multiple meanings, the definitions presented here are specific to the context of this article.

Hawaiian	***English***
ahupua‘a	Land division usually extending from the uplands to the sea; the landlord or owner of an ahupua‘a might be a konohiki
‘āina	Land, earth, to feed
‘auamo	Pole or stick used for carrying burdens across the shoulders
aloha	Love, affection, compassion, mercy, sympathy, pity, kindness, sentiment, grace, charity, greeting, salutation, regards, sweetheart, lover, beloved, loving, kind, compassionate, charitable, loveable
Aloha ‘āina	Love of the land or one’s country, patriotism, nationalism
hoa‘āina	Tenant, caretaker, as on a kuleana
hō‘ailona	Sign, symbol, representation, omen
haumāna	Student, pupil, apprentice, recruit
i‘a	Fish or any marine animal, as eel, oyster, crab, whale
‘Ike	to see, know, feel, greet, recognize, perceive, experience, be aware, understand, knowledge, comprehension, learning, sense, vision

kāhea	to call, cry out, invoke, to summon; a call, alarm
Kahuna	priest, sorcerer, expert in any profession
kalo	Taro, staple food of the Kanaka ʻŌiwi
Kanaka ʻŌiwi	Native Hawaiian, Native born, Indigenous
kapu	Taboo, prohibited, special privilege, sacred, forbidden
keiki	Child, offspring, descendant, progeny
kiaʻi loko iʻa	Fishpond caretakers, guardians
kilo	To watch closely, observe, forecast
kōkua	Help, aid, assistance, relief, assistant,
konohiki	Land agent, Headman of an ahupuaʻa land division under the chief; land or fishing rights under control of the konohiki; such rights are sometimes called konohiki rights
kuleana	Right, privilege, concern, responsibility, reason, function
kūpuna	Ancestors, grandparents, relatives or close friends of the grandparent's generation
kupuna	Singular of kūpuna (above)
lawaiʻa	Fisher; fishing technique; to fish, to catch fish
limu	Seaweeds; a general name for all kinds of plants living under salt and freshwater
loina	Rule, customs, manners, code, precept, law, principle, statute, ordinance
loko iʻa	Fishpond
makai	Ocean, seaward, toward the sea
mākua	Parent, any relative of the parents' generation, as uncle, aunt, cousin
mana	Supernatural or divine power, authority; to give mana to, to make powerful; to have mana, power, authority; authorization, privilege; miraculous, divinely powerful, spiritual; possessed of mana, power.
mana moi	Mature male life stage of moi
mauka	Inland, direction opposite to the sea
Mohala Nā Konohiki	Mohala Nā Konohiki is an initiative that was co-created by Uncle Mac Poepoe and Kēhau Springer and catalyzed by Conservation International to support the blossoming (mohala) of the next generation of lawaiʻa and konohiki apprentices.
moi	Threadfish *Polydactylus sexfilis*
moi liʻi	Small threadfish, juveniles, first stage of moi typically under six inches
moku	District, island, islet, cut, severed, broken

moʻokūʻauhau	Genealogy
naʻau	Intestines, bowels, guts, mind, heart, affections; of the heart or mind; mood, temper, feelings
Nā Kai ʻEwalu (NK8)	The eight oceans was a name given to our group to honor our connection to all the oceans and to demonstrate how the oceans and currents are like a lei that string all of us together.
nūpepa	Hawaiian language newspapers
ʻohana	Family, relative, kin group, related
ʻōpelu	*Decapterus pinnulatus*
ʻōpio	Youth, juvenile, youngster, junior
Pae ʻĀina o Hawaiʻi	Hawaiian islands, archipelago
pala Moi	Hermaphroditic growth stage of the moi
pilina	Association, relationship, union, connection
pono	Goodness, upright, morality, proper, equitable
Wahi kūpuna	Ancestral sites, location, place
wahine	Woman, lady, wife

Supplementary 2: Examples of *Loina lawai‘a* Kanaka ‘Ōiwi Fishing Customs.

These customs apply to all species, including moi, and provide important context for how Hawaiian Traditional Practice is carried out for learning about and stewarding moi.

Loina	**English Translation**
E ‘ai i kekahi, e kāpī i kekahi	Eat and harvest until you are fulfilled; Take only what you can eat, with concern for the wellbeing of future generations and ability to meet present food needs without compromising the ability of future generations to meet their needs.
Lōkahi	Time spent fishing cultivates intimacy and harmony with the ocean and understanding environmental processes. It also strengthens relationships with marine creatures that are a part of Hawaiian identity and spirituality
Mālama	To nurture and respect all living things and the relationships between kanaka and the environment.
Ha‘aha‘a	Humility: we are a part of a living world and not superior to it. By excluding people from nature only serves to further alienate humans from other living resources and thus from their responsibilities.
Kuleana	Responsibility; assume ownership for your actions when interacting with the environment.
‘Imi ‘ike	Seek knowledge; the transfer of knowledge from one generation to the next by training apprentices on the subtle and major conditions of the marine resources and environment.
E ho‘iho‘i i ka i‘a mua	Return the first fish caught; an act of reciprocity with the intention of giving back to the ocean to sustain resource availability in perpetuity.
Mai‘a	Banana; Don’t bring bananas to the ocean “Inā he moe mai‘a, makehewa e hele i ka lawai‘a”
Pa‘a ka waha	Close the mouth; don’t talk when fishing to avoid startling fish.
Mai ‘a‘e	Don’t step over your work; respect your fishing implements
Holoholo	To cruise; a term used to conceal your fishing plans from others and avoid scaring fish who will know your intentions resulting in failure.
Mai kuhikuhi	Don’t point at a fish you’ve sighted to avoid drawing its attention.
‘A‘ole nui ka ukana	Don’t bring too many things to the ocean to better focus your efforts on a specific target species.

Supplementary 3: Dedication to the late Kanohowailuku Helm

We sit in gratitude for ALL the kūpuna, lawai‘a, konohiki and aloha ‘āina who have paved the way so that we may continue this work. We are humbled to be a single ‘aho, a strand in the weaving of knowledge, ‘ike and mana that binds our past to our future. This research and publication is dedicated to Kanohowailuku (Kanoho) Helm - son of Adolph and Corene Kēhaunani Helm and nephew of aloha ‘āina leader, George Helm - whose lives embody the enduring spirit of aloha ‘āina, lawai‘a wisdom and Molokai pride.

Kanoho Helm was the beloved kāne of Lindy Helm and proud father of Kealohanui, Kekumu and Kaua Helm. A fisher, farmer, paddler, singer-songwriter and cherished friend, he inspired so many of us with his aloha, charisma and dedication to his ‘ohana, ‘āina and lāhui. A true son of Molokai, he spoke fearlessly for what was pono, standing for ‘āina, kai and our kūpuna with humility and conviction. Kanoho’s brilliance as a “kanaka scientist” placed him far ahead of his time. His understanding of the ocean and its rhythms surpassed data or degrees; he connected to the realms of the seen and unseen, communicating with the natural world through deep ancestral knowledge. Among his greatest gifts was his ability to teach and embody the principles of traditional Hawaiian resource care. From his early days at Mo‘omomi, he trained as a lawai‘a dedicated to pono practice and stewardship. His voice and presence bridged generations, reminding us that aloha ‘āina is lived through action - with every throw of the net and every respectful step upon our shoreline.

Kanoho was also a gifted poet and songwriter whose music carried the principles of aloha ‘āina and resource care. His album, nominated for a Nā Hōkū Hanohano Award (2012), revealed his rare ability to blend melody with message, as heard in songs like *Lei Alana ‘Opihi Mama Cellana*, where his words honor the intimate relationship between people, place and sea. Through his compositions, teachings, and aloha, his legacy continues to guide us towards balance and kuleana and deeper connection to ‘āina. E ku‘u hoa ē, pau ke kaumaha, ke kui ana ka lei Poepoe ma Nā Kai ‘Ewalu, a noke mau, e hele ‘oe, e hele a e moe mālie.

Some of his words that continue to inspire and carry forward:

“Please don‘t change Molokai, let Molokai change you…Be a part of the solution”
- I Aloha Molokai (IAM)

“We were born into the sea. E ho’i mai, e Lonomakahiki. We were taught to love this land. It’s not so hard to understand.” Lyrics: Homestead Farmer

“What would you do with a billion dollars? I‘d buy a fish, a billion dollar fish…..”
Lyrics: Billion Dollar Fish

Supplementary 4: Hawai‘i Extreme Weather Events (2014-2024).

This table documents extreme weather events affecting Hawai‘i, providing a holistic view of environmental observations starting five years prior to our documented out-of-season moi spawning. It offers valuable insights into long term climate fluctuations and their potential impacts on moi and other culturally important species.

Year/ Month	Location	Event	Environmental Impacts	Potential Impacts on Moi
2014 - Aug	Hawai‘i Island - Southeast, Statewide	Hurricane Iselle made landfall as a Tropical storm on Southeastern part of Hawai‘i Island	Flash flooding increased sedimentation in nearshore habitats and reduced water quality	Increased turbidity influences foraging success and elevates exposure to terrestrial based pollutants, pathogens and parasites from storm water discharge into moi habitats
2014 - 2015	Statewide	El Nino - strong	Elevated seawater temperatures caused two consecutive coral bleaching events during the summers of 2014 and 2015 throughout the Hawaiian archipelago resulting in widespread loss of live corals in shallow reefs	Shifts in ocean current strength and direction influences larval recruitment and population connectivity patterns. Physiological stress from exposure to elevated seawater temperature affects sensory and reproductive development, growth rates, spawning behavior and cycles
2018-2020	O‘ahu - Kawahaokamanō	Invasive Limu Die Off	Reef and estuary habitats available for colonization by other benthic species	The reef and mudflats were clean, freeing the substrate of invasive algae. Honu (green sea turtles) were observed hungry and starving and forced to forage very close

				to shore
2016 - Jul	Hawai‘i Island - Southeast	Hurricane Darby made Landfall on southeast of Hawai‘i Island	Flash flooding increased sedimentation in nearshore habitats and reduces water quality	Increased turbidity influences foraging success and elevates exposure to terrestrial based pollutants, pathogens and parasites from storm water discharge into moi habitats
2016 - July to 2017	Hawai‘i Island - Puna	Kamokuna lava flow reaching the ocean	Elevated seawater temperature, widespread marine species mortality and extirpation of coral reef habitat	Habitat destruction and widespread mortality of marine species (covering of poho moi “moi holes” by lava) Warmer waters may also alter growth rate and spawning cycles
2017 to present	Statewide	King Tides and Coastal Flooding	Seawater inundation into coastal environment increases urban runoff and salinity levels	Altered salinity causes stress on moi, impact to spawning habitat
2018 - Apr	Kaua‘i - Ha‘ena	Tropical Storm Lane	Flash flooding increased sedimentation in nearshore habitats and reduces water quality	Increased turbidity influences foraging success and elevates exposure to terrestrial based pollutants, pathogens and parasites from storm water discharge into moi habitats
2018 - May	Hawai‘i Island - Puna	Lower East Rift Zone eruption, fissures opening in Leilani Estates w/ multiple ocean	Elevated seawater temperatures, widespread marine species mortality and extirpation of coral reef habitat	Habitat destruction and widespread mortality of marine species (covering of poho moi “moi holes”

		entries		by lava) Warmer waters may also alter growth rate and spawning cycles
2018 - Aug	Hawai‘i Island, Maui	Hurricane Lane	Flash flooding increased sedimentation in nearshore habitats and reduced water quality	Increased turbidity influences foraging success and elevates exposure to terrestrial based pollutants, pathogens and parasites from storm water discharge into moi habitats
2018 - Sep	Maui, Lāna‘i	Tropical Storm Olivia	Flash flooding increased sedimentation in nearshore habitats and reduced water quality	Increased turbidity influences foraging success and elevates exposure to terrestrial based pollutants, pathogens and parasites from storm water discharge into moi habitats
2018 - Sep	Lalo, FFS Papahānaumokuākea	Hurricane Walaka - Cat 3 Completely destroying East Island	Flash flooding increased sedimentation in nearshore habitats and reduced water quality	Increased turbidity influences foraging success and elevates exposure to terrestrial based pollutants, pathogens and parasites from storm water discharge into moi habitats
2019	Statewide	El Nino - strong	Elevated seawater temperatures caused coral bleaching throughout the Hawaiian archipelago resulting in loss of live coral cover in shallow reef habitat	Shifts in ocean current strength and direction influences larval recruitment and population connectivity patterns.

				Physiological stress from exposure to elevated seawater temperature affects sensory and reproductive development, growth rates, spawning behavior and cycles
2019-2020	Statewide	“The Blob” Marine Heatwave	Elevated seawater temperatures caused two consecutive coral bleaching events during the summers of 2014 and 2015 throughout the Hawaiian archipelago resulting in widespread loss of live corals in shallow reefs	Shifts in ocean current strength and direction influences larval recruitment and population connectivity patterns. Physiological stress from exposure to elevated seawater temperature affects sensory and reproductive development, growth rates, spawning behavior and cycles
2020 - April	O‘ahu, Maui	COVID-19 Pandemic	Reduction in disturbances for tourism overuse allowed for revitalization of valued resource species of algae and fish including limu līpoa (*Dictyopteris plagiogramma*) and nenue (*Kyphosus bigibbus*)	Increased fishing pressure and moi mortality due to a general rise in leisure engagement by recreational anglers, coupled with decrease in overall tourist visitorship which reduced anthropogenic disturbances unrelated to fishing in moi habitats
2021 - Dec	Statewide	Kona Low Storm	Flash flooding increased sedimentation in nearshore habitats and reduced water	Increased turbidity influences foraging success and

			quality	elevates exposure to terrestrial based pollutants, pathogens and parasites from storm water discharge into moi habitats
2022 - Nov	Hawai‘i - Mauna Loa,	Eruption contained to high elevations	Increase in sulfur dioxide, and vog affects air quality, resulting in acidification reducing seawater pH levels	Habitat destruction and widespread mortality of marine species (covering of poho moi “moi holes” by lava) Warmer waters may also alter growth rate and spawning cycles
2022 - Dec	Hawai‘i Island, Maui, O‘ahu	Tropical Storm Darby	Flash flooding increased sedimentation in nearshore habitats and reduced water quality	Increased turbidity influences foraging success and elevates exposure to terrestrial based pollutants, pathogens and parasites from storm water discharge into moi habitats
2023 - Jan	Moloka‘i	Tropical Storm Calvin	Flash flooding caused erosion and silt runoff into loko i‘a and estuaries, increasing turbidity in nearshore marine habitats	Increased turbidity influences foraging success and elevates exposure to terrestrial based pollutants, pathogens and parasites from storm water discharge into moi habitats
2023 - Aug	Maui	Hurricane Dora passed south of islands producing strong winds	Flash flooding increased sedimentation in nearshore habitats and reduced water quality	Increased turbidity influences foraging success and elevates exposure to

		which caused Fire Storm in Lāhainā		terrestrial based pollutants, pathogens and parasites from storm water discharge into moi habitats
2024 - Aug	Hawai‘i	Hurricane Hone	Flash flooding increased sedimentation in nearshore habitats and reduced water quality	Increased turbidity influences foraging success and elevates exposure to terrestrial based pollutants, pathogens and parasites from storm water discharge into moi habitats

Supplementary 5: Adaptive Management Tools and Options gathered by the Nā Kai ʻEwalu moi collective.

Management Tools	**Example**	**Outcomes**
Changes to Sampling Permitting for CBSFAs *Fast-Track option – faster processing* *Blanket permit option – no need for renewal*	Marine Corps Base Hawaiʻi *Blanket permit* Hawaiʻi Institute of Marine Biology *Fast-track permit*	CBSFAs are able to conduct key research for stewarding resources without having to go through lengthy permit processes.
Lawaiʻa data respected as the basis for management adjustments	Miloliʻi CBSFA Kīpahulu CBSFA Hāʻena CBSFA	Lawaiʻa given respect as co-managers for marine areas; lawaiʻa data and insight used to determine management policy.
Marine management plan (including moi)	Kīpahulu CBSFA	Plan can weigh the needs of the local people and ecosystem e.g subsistence needs, cultural gathering needs, fish population, pollution, tourism
Change kapu months (currently June-Aug)	Moʻomomi CBSFA Proposed Rules Kīpahulu CBSFA *Kapu from May-September*	Increased chance that moi spawning will be successful, resulting in more offspring for future generations.
Slot limits instead of minimum size	Kīpahulu CBSFA *During the open season less than 11 inches or more than 18 inches long is kapu or prohibited from gathering.*	Increased chance that moi spawning will be successful, resulting in more offspring for future generations.
Kapu largest females	Kīpahulu CBSFA	Moi of larger sizes (fecund females) are able to spawn more and create more viable offspring to replenish the next generation.
Limit disturbance in nursery grounds	Hāʻena CBSFA Moʻomomi CBSFA Proposed Rules Kawaʻaloa Bay Nursery	Moi liʼi will be able to feed and shelter, thus increasing the rate of survival into large reproductive moi.
Restrict fishing gear	Moʻomomi CBSFA Proposed Rules Moi can only be taken by hook-and-line, spear, or throw net	To promote traditional and subsistence fishing practices.

Supplementary 6: Qualitative Indications of Moi Spawning.

Sex	Gonad Characteristics	Observation
Female	Color & quality	Orange / yellow, bright color, grainy, inflated
Female	Size	Large relative to the size of the fish
Female	Presence of Veins	Red, thick, visible
Female	Presence of Eggs	Visible immediately, or when a small amount of pressure is applied
Female	Egg quality	Large and hydrated, inflated, eggs visible by eye
Male	Color & quality	White / cream / pink, bright color, smooth, inflated
Male	Size	Large relative to the size of the fish (always thinner than the gonads)
Male	Presence of Veins	Rarely, but if so this is also a positive indication
Male	Presence of Sperm ‘milting’	Visible immediately, or when a small amount of pressure is applied

Supplementary 7: Parameters for quantitative data analysis of Moi Spawning Period and Moi Size-at-Sex.

Question	**Null Hypothesis H0**	**Source**	**Distribution modeling**	**Mean and Spread of expected data.**	**Assumed spread (standard deviation) of published ranges**		
					1 stdev *Least conservative*	**2 stdev** *Middle*	**3 stdev** *Most conservative*
Do moi spawn *outside the known period*?	Moi spawn from June 1 to August 31. Spawning outside this period is negligible.	Hawai'i State Regulations	Frequency of spawning modeled as a Normal distribution	Midpoint of the spawning season under H0 is July 31st (Julian day 212). The spread is 92 days.	σ=46	σ=23	σ=15.3
Are moi becoming *males* at larger or smaller sizes?	Moi sex organs change to male between 200-250mm (50mm range). Outside of this size range fish are not males.	Poepoe et. al 2001 (12)	Range of male moi sizes modeled as a Normal distribution	Midpoint of male moi sizes under H0 is 225mm. The spread is 50mm.	σ=25	σ=12.5	σ=8.3
Are moi becoming *females* at larger or smaller sizes?	Moi sex organs change to female from 300-400mm (100mm range). Under this size range fish are not female. Over this size range all are female. 610mm is the largest recorded moi.	Poepoe et. al 2001 (12)	Range of female moi sizes modeled as a Normal distribution	Midpoint of female moi sizes under H0 is 350mm. The spread is 100mm.	σ=50	σ=25	σ=16.7

Supplementary 8: Detailed Approach for Quantitative Data Analysis (Moi Spawning Period and Moi Size-at-Sex)

We performed statistical analyses using RStudio version (2025.05.1+513) and R version R 4.5.1. We used the R base package for calculations alongside the dplyr and ggplot2 packages for data manipulation and visualization.

Moi Spawning Period

Null Hypothesis (H0): Current State of Hawai'i regulation assumes that the vast majority of spawning activity occurs within the protected June-August window. Spawning activity outside this window is negligible or non-existent. **We aimed to compute the probability of observing our sample results based on H0.**

Step 1: Find the Probability (p) of observing spawning fish (our data) outside the protected period from June 1 to August 31. For the mean, we chose the midpoint of the spawning season under H0, July 31st, which is the 212th day of the year (μ = 212). We modeled three scenarios where the published range (i.e the 92 days from June 1 to August 31) comprised either 1, 2, or 3 standard deviations from the mean.

Step 2: Find the probability of observing (k) spawning fish out of (n) sample. In some cases we observed that only a fraction of fish were spawning capable at a particular date. To account for this we used the Binomial Probability Formula P(X=k) = C(n, k) * p^k * (1-p)^(n-k) to compute the probability of observing our specific data, where n is the total number of fish in our sample; k is the number of spawning fish observed, and p is the probability of a single fish spawning in October (from Step 1). C(n, k) is the number of combinations ("n choose k").

Moi Size at Sex

Null Hypothesis (H0): Previous studies record that male moi mature from 200-250mm and female moi from 300-400mm. **We aimed to compute the probability of observing our sample results based on H0.**

Step 1: Find the Probability (p) of observing males and females (our data) outside the recorded ranges (200-250mm for males and 300-400mm for females). For the mean, we chose the midpoint of the size range for males and females under H0, for males (μ = 225) and females (μ = 350). We modeled three scenarios where the published range (i.e 50mm for males and 100mm for females) comprised either 1, 2, or 3 standard deviations from the mean.

Step 2: We compared the mean and observed fork lengths (males and females treated separately) to calculate a Z-score (absolute), to which we applied the probability density function for the normal distribution. From this result we derived the two-tailed p-value for the moi sizes from each sampling trip.

Step 3: Linear mixed models

We modeled fork length using a normal (gaussian) distribution since this distribution was created to model continuous data such as height and length, and models from this distribution were well fit, resulting in a closely matched QQ plot, and without indication of over or under dispersion, and no warnings according to DHARMa package model fitting tests.

Our fixed effects were: Sex (factor), Island (factor), Year (factor) and Month (integer). Our interaction effects were: Sex*Island, Sex*Month and Island*Month. Our data coverage was not appropriate to model a 3-way interaction effect. Models with different combinations of variables were selected by keeping any effects that contributed significantly to explaining fork length variation, and according to AICC, which is a model selection metric suitable for data with small sample sizes.

Ultimately the selected model for interpretation included all possible fixed and interaction effects. The AICc for this model was 1174.7. The r-squared equivalent was manually calculated using an R-modified version of the python code shared here (https://stats.stackexchange.com/questions/477598/equivalent-of-r-squared-in-generalized-linear-model-regression-results) and resulted in 0.7504138 for this model. We used a chi-squared test (drop1(model, test = "Chisq")) to examine whether removing any of the model fixed or interaction effects would impact the model fit. We used the function eemeans() to determine the mean and upper and lower confidence levels for Male and Female moi fork lengths across different islands.

Additionally, we fit linear models to test the specific hypotheses of whether (1) Female moi on Maui were smaller than on other islands and (2) Male moi on O'ahu were larger than on other islands. Taking this approach allowed us to model the temporal variables of month and year as random effects. Month was modeled as a factor instead of an integer to more accurately reflect that months are not linear and the 'first' month follows the 'twelfth'.

Our statistical approach was reviewed by Etaash Katiyar (Stanford University), Wesley Bian (Stanford University), Louis Reeve Stanley Davis (Stanford University) and Jules Wyman (Washington University). For full details please see the RMarkdown file in Supplementary 9.

Supplementary 9: R-markdown file for Quantitative Data Analysis (Moi Spawning Period and Moi Size-at-Sex).

File type: RMarkdown file

Required: RStudio version (2025.05.1+513) and R version R 4.5.1. and Input and Output data files which are provided in the Supplementary 10 excel workbook. The sheets of the workbook should be downloaded as .csv files to run with the code.

Supplementary 10: Data Tables and Visualization for Quantitative Data Analysis of (a) Moi Size-at-Sex and (b) Moi Spawning Period.

File type: Excel workbook

Required: Microsoft Excel software.

Supplementary 11: Author Introductions and Group picture.

KŪPUNA (Grandparent Generation)

Kelson "Uncle Mac" Poepoe
I am a lawaiʻa from Molokaʻi. I fish the way my kūpuna taught me. Families like mine feel a sense of responsibility to steward what the ocean gives us. Our actions are based on kilo – careful observations. I am happy to pass on my knowledge to the next generation. But for that sharing of knowledge to take place, there first has to be respect for indigenous knowledge. Over the past 40+years, Iʻve been working with the next generation of lawaiʻa and stewards from across Hawaiʻi because to pass these traditions and to inspire lawaiʻa pono (proper, moral, equitable).

Keliʻi Alapaʻi
I am a lawaiʻa from the ahupua'a of Hāʻena, Kaua`i. Raised and taught the traditional practices used by kūpuna and the late Thomas Hashimoto, who was the last of the master lawaiʻa from Hāʻena. My ʻohana lives off the land and sea. We need to pass on our ʻike to our children so they can carry on our gathering traditions and knowledge that was passed on to us.

MĀKUA (Parent, Aunt and Uncle Generation)

Kēhau Springer
No Kaimukī, Oʻahu mai au akā noho wau me kuʻu ʻohana ma Keaʻau, Puna, Hawaiʻi (Iʻm from Kaimukī, Oʻahu but I live with my family in Keaʻau, Puna, Hawaiʻi). Iʻve been fortunate to work in ocean conservation for over 20 years and have had the honor of collaborating with Uncle Mac Poepoe for 15+ as his student helping to develop programs that promote traditional Hawaiian resource management. My doctoral research focuses on Kanaka ʻŌiwi representation in Hawaiʻi's ocean governance.

Hiʻilei Kawelo
I am a fifth generation wahine (woman) lawaiʻa from Kawahaokamanō (Kāneʻohe Bay) and 25 years as a kiaʻi loko (fishpond guardian/caretaker) of Heʻeia Fishpond. My ʻohana roots are in Kaʻalaea and Kahaluʻu ahupuaʻa and grew up learning to gather and harvesting all different kinds of ocean resources from our bay.

Nākoa Goo
I am a marine scientist and lawaiʻa from Puna, Hawaiʻi currently working at the Hawai'i Institute of Marine Biology. My doctoral research is focused on foraging ecology and the cultural significance of native forage fishes in estuaries throughout Hawaiʻi. As a member of the Akau ʻohana (family), my heritage is strongly linked to the Kawaihae region of Hawaiʻi island. I currently collaborate with communities throughout Hawaiʻi to support projects focused on management and stewardship of marine resources.

Pekelo Lind
ʻO Pekelo Lind koʻu inoa. He noho papa au ma Kīpahulu, Maui, Hawaiʻi me koʻu ʻohana. I was fortunate to grow up under the wings of my late father, John Kimokeo Lind, whose moʻolelo, lessons, and mastery, molded me into who I am today - a lifelong subsistence lawaiʻa, hunter, gatherer, and farmer. At a young age, I learned the importance of kilo and began applying it to almost every aspect of my life. Through the practice of kilo, I was able to notice changes in our nearshore species and document place-based spawning periods of specific and significant species that inspired some of the rules in the Kīpahulu Community-Based Subsistence Fishing Area. Uncle Mac Poepoe has deepened my understanding about

kanaka science when it comes to the moi lifecycle, and how we can use this science to understand the various changes occurring. I am passionate about raising awareness of place-based kilo and monitoring, grounded in indigenous knowledge, to inspire and empower individuals across our pae‘āina to make a meaningful difference in their communities.

Kāneholani Lind
I am a lineal descendant of Maui Hikina, Hawaiian practitioner, subsistence fisherman, kalo farmer, hunter, father, and community member in the rural village of Kīpahulu. I have inherited knowledge from my father, John Kimokeo Lind, the late Konohiki of Kīpahulu Moku, inspired to mālama our fish and ocean resources in perpetuity.

Will Ma‘e-Huihui
My name is William Ma‘e-Huihui, born in Hilo raised in a commercial fishing family. I fished from Pohoiki, Hilo, Kawaihae, Kona, all the way to Miloli‘i. I learned the fishing practices of each community through ‘ohana ties to the area. As I got older I noticed a decline in fishing and wanted to help make a difference to mālama our oceans. I am honored to have met such amazing people in my journey that want to help replenish our ocean resources. Under the tutelage of uncle Mac Poepoe and Mohala Nā Konohiki (apprenticement program) I was able to join these amazing lawai‘a to share our collective knowledge, with this Moi Collective ohana I have learned not only more about Moi and their life cycle but also their correlation between the other Hawaiian Islands.

Laila Kaupu
‘O wau o Laila Kaupu. No Omoka‘a mai au. Noho au ma Ho‘ōpūloa-Miloli‘i. I am Laila Kaupu. I‘m from Omoka‘a. I live in Ho‘ōpūloa-Miloli‘i. I am a 5th generation ‘ōpelu (*Decapterus pinnulatus*) lawai‘a practitioner of Miloli‘i, a steward of ‘ohana ‘āina (family land) Omoka‘a, an apprentice for Mohala Nā Konohiki Miloli‘i, and haumāna (student) of Nā Kai ‘Ewalu both under the keen guidance of Uncle Mac Poepoe.

Makani Gregg
I am originally from Kailua and Waimānalo, O‘ahu and now living in Pohoiki, Hawai‘i. I am aloha ‘āina (patriot/practitioner that acts upon love of land) and practice all different things mauka (inland) to makai (ocean). My community that I mostly serve is lower Puna, Pohoiki to Kalapana but I do kōkua (help) in other communities across Hawai‘i.

Kanoe Alapa‘i-Hashimoto
Aloha, I am born and raised from Kaua‘i, moku (district) of Halele‘a, Hā‘ena Ahupua‘a. Grandson to the late Thomas Hashimoto and student of Hawaiian traditional gathering practices.

Avery Kaho‘okaulana
I was born and raised from Kapoho to Kalapana, Hawai‘i. I enjoy food harvesting/gathering and processing mauka to makai. I also serve the lower Puna community now from Pohoiki to Kalapana because of the lava, we don’t go to Kumukahi and Kapoho as much anymore.

Alex Connelly
I‘m from Ko‘olau Poko, O‘ahu and have worked for Kua‘āina Ulu ‘Auamo since 2011 to support E Alu Pū, a learning network and movement for community-based natural resources management that Uncle Mac founded in 2003.

‘ŌPIO (Youth Generation)

Marina Dewinara Luccioni

I am from the islands of Corsica and Java, and grew up in London and Corsica. I live in Half Moon Bay, California. I have been mentored by Uncle Mac Poepoe since 2018 and am a PhD candidate studying marine chemical ecology, advised by Larry Crowder at Stanford University.

